

Hierarchical Fast–Slow ReAct Agent for Zero-Shot Object-Goal Navigation

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Abstract

Zero-shot object-goal navigation (ZSON) requires a robot to find a named object category in a building it has never entered. The prevailing approach scores frontiers with a vision–language *value map*: every decision is another argmax over the map as it currently stands, and the evidence behind that score is discarded the moment it is taken. Systems that place a large vision–language model inside the perception–action loop typically query it on a fixed schedule from the current view alone; a room the robot walked through minutes earlier is never reconsidered, and a failed call has no defined fallback. We turn what the robot has already seen into the object of deliberation. Our hierarchical fast–slow agent leaves the value-map controller running at every step and writes a *coordinate-anchored memory* as it moves: a semantic grid of room types and confirmed object instances, together with a bounded store of pose-tagged keyframes. A VLM screens each candidate detection before it is written. A deliberative layer reads this memory in a bounded reason–retrieve–act loop. It wakes on structural events the reactive layer computes, reasons first over text, and recalls a first-person view only for candidates that text alone cannot separate. Per-invocation and per-run caps bound its calls, a call-free first tier resolves the most frequent stall, and any failure returns control to the reactive controller. Our system reaches 68.75% SR on HM3D v1 val and 47.29% on MP3D val, the highest success rate among the zero-shot methods compared here. Choosing among far frontiers by argmax instead of deliberating costs 3.40 SR points in a paired comparison over all 2000 HM3D episodes (95% CI [1.70, 5.05]); deliberating over every frontier does not recover them.

1 Introduction

Zero-shot object-goal navigation (ZSON) places a robot at a random pose in a previously unseen indoor environment and asks it to find an instance of a named object category—*bed*, *toilet*, *plant*—without any category-specific training (Batra et al. 2020; Majumdar et al. 2022). A prominent family of solutions explores reactively, scoring each frontier between explored and unknown space with a vision–language *value map* that measures how much the view toward it resembles the goal (Yokoyama et al. 2024; Zhou et al. 2023; Yu, Kasaei, and Cao 2023; Gong et al. 2026).

What such a controller lacks is a way to act when it has no action it believes in: clutter defeats the local planner, the only frontiers left lie far away, the frontier set runs out because an opening was missed, or the goal sits on another floor. These are *structural impasses*—cheap for the reactive layer to detect, and beyond what another argmax over its current value map can resolve. Three of the four turn on an earlier observation the controller no longer keeps. Systems that bring large language and vision–language models into navigation (Cao et al. 2025; Yin et al. 2024) call the model on a schedule fixed before the robot moves, and hand it evidence fixed with that schedule: a room the robot walked through minutes earlier cannot be re-examined, since a summary of it may survive but the view itself does not. And how that consultation is governed—when to call, how much evidence to gather, where control goes when the call fails—is left unspecified.

Our agent is hierarchical and fast–slow (Fig. 1): a reactive value-map controller runs at every step and, as it explores, writes a *coordinate-anchored memory*—a semantic grid of room types and confirmed object instances, plus a bounded store of pose-tagged keyframes. A goal detection—the only evidence that can authorize stopping—passes a verification gate that vetoes it *before* fusion makes it permanent. A bounded reason–retrieve–act loop (Yao et al. 2023) reads that memory: it wakes on the structural events the reactive layer computes, reasons first over text, and pulls up a remembered first-person view only for candidates that text alone cannot distinguish. Two-level budgets cap every loop, and every failure path returns control to the reactive layer, which stays available until it reaches its own terminal condition.

On HM3D v1 and MP3D val (Ramakrishnan et al. 2021; Chang et al. 2017) the full system leads every zero-shot baseline of Table 1. Choosing among far frontiers by argmax instead of deliberating costs 3.40 points of SR, while deliberating over every frontier recovers none of that loss.

Our contributions are:

1. **A hierarchical fast–slow agent for ZSON.** A reactive value-map controller keeps the robot moving at every step, while a deliberative ReAct layer sits off that per-step path and intervenes only at structural impasses: deliberation redirects the robot; it is never what drives it.
2. **A coordinate-anchored memory with an abstract and a literal tier.** A semantic grid accumulates room types

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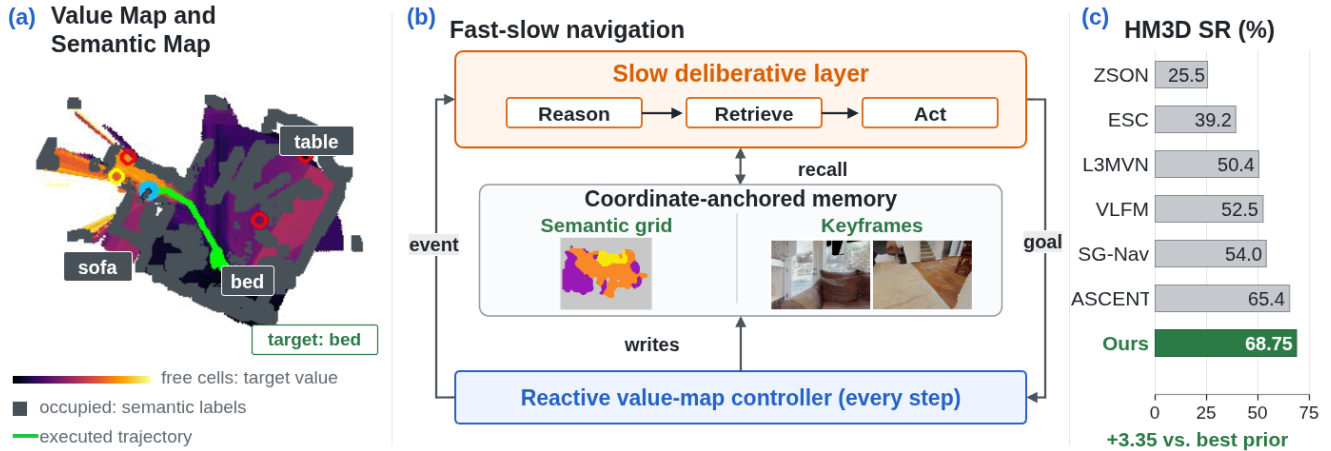


Figure 1: **Hierarchical fast-slow navigation at a glance.** (a) One HM3D run with *bed* as the goal: the reactive value-map controller scores free cells by how much the view toward them resembles the goal (colour bar), labels occupied cells as it recognises them, and executes the green trajectory. (b) That controller runs at every step and writes a coordinate-anchored memory—a semantic grid and a store of pose-tagged keyframes. A bounded reason–retrieve–act loop wakes on the structural events the controller reports, recalls what it needs, and returns a new goal. (c) The full system reaches 68.75% SR on HM3D v1 val, the highest among the zero-shot methods we compare.

and object instances over the whole run; a bounded store keeps the pose-tagged views themselves; one coordinate frame indexes both, so the deliberative layer reasons over text first and opens the view behind a claim only when text cannot decide. Since neither tier can undo a fused observation, each admits evidence by what that evidence can cause: an instance becomes retrievable after two sightings, and a goal detection must pass a per-detection verification gate before fusion.

3. **An invocation policy that makes deliberation bounded and fail-open.** When to call, how much evidence one call may gather, and where control goes when it fails are parts of the method rather than deployment details: the trigger is a structural impasse the reactive layer already computes, a call-free first tier answers the most frequent stall, per-invocation and per-run budgets cap the rest, and every failure path returns control to the reactive controller.

2 Related Work

Zero-shot object-goal navigation The ObjectNav task definition and evaluation protocol were consolidated by Batra et al. (2020), with progress measured in the Habitat simulator (Savva et al. 2019) on the HM3D (Ramakrishnan et al. 2021) and Matterport3D (MP3D) (Chang et al. 2017) scene datasets, where frontier-based exploration (Yamauchi 1997) and learned semantic-map policies (Chaplot et al. 2020) set the early benchmark. The zero-shot line removes category-specific training: CoW (Gadre et al. 2023) drives frontier exploration with an open-vocabulary detector; ZSON (Majumdar et al. 2022) transfers image-goal training through multimodal goal embeddings; ESC (Zhou et al. 2023) and L3MVN (Yu, Kasaei, and Cao 2023) inject language-model commonsense into frontier selection; VLFM (Yokoyama et al. 2024) scores frontiers directly with

a vision–language value map; and ASCENT (Gong et al. 2026) extends the value-map recipe with floor-aware, coarse-to-fine exploration. All of these controllers are reactive: each decision consumes the current map and observation through a fixed scoring rule, and when that rule reaches a structural impasse there is no deliberative recourse. Nor is there a check between detecting a goal candidate and writing it into the map: VLFM and ASCENT fuse what the detector reports.

LLM/VLM-guided navigation and scene-graph reasoning A second line consults a large model during navigation. CogNav (Cao et al. 2025) models the navigation process as transitions among cognitive states orchestrated by an LLM; SG-Nav (Yin et al. 2024) incrementally builds an online hierarchical 3D scene graph and prompts an LLM with it for zero-shot object navigation. In both, *when* the model is called is settled before the robot moves—every step, or on a prespecified sequence of states—and every call is handed the same kind of evidence. What the robot saw earlier reaches the model only as whatever its representation chose to keep—SG-Nav’s graph nodes, or, in image-goal navigation, the trajectory fragments MemoNav (Li et al. 2024) judges informative—never as the view itself. Spatial memories built for language queries—VLMaps (Huang et al. 2023) fusing vision–language features into a 3D map, Concept-Graphs (Gu et al. 2024) into an open-vocabulary 3D scene graph—keep descriptors of a place rather than the views of it, and neither decides when a model should be consulted. Our controller instead picks its call times at run time, from four structural impasses the reactive layer detects as it runs (Section 3.4).

ReAct agents A reason–act agent interleaves reasoning, tool calls, and the observations those calls return, so the

model itself decides at run time what to examine next (Yao et al. 2023). Agents of this shape now close real GitHub issues (Jimenez et al. 2024; Yang et al. 2024) and run chemistry experiments on laboratory hardware (Boiko et al. 2023; Bran et al. 2024). Embodied work has adopted the loop by handing it the run: ORION (Dai et al. 2024) emits thought-action pairs over perception and memory tools to drive a Habitat robot through dialogue-driven personalized search; TANGO (Ziliotto et al. 2025) has an LLM compose navigation and perception primitives into a program written once, before the robot moves; and CogNav (Cao et al. 2025) re-queries the model at every decision to advance a cognitive state machine. We put the loop beneath the reactive controller instead of in front of it: the controller drives every step unaided, the loop wakes only at a structural impasse, and the model then chooses for itself which part of the anchored memory to read. Section 4.3 contrasts the two placements directly.

3 Method

We address zero-shot object-goal navigation (Batra et al. 2020): a robot placed at a random pose in an unseen, multi-room, possibly multi-floor building must find an instance of a category g . At each step it receives an egocentric RGB-D observation and an odometry pose $\mathbf{p}_t = (x_t, y_t, \psi_t)$, and emits a discrete action.

3.1 Architecture Overview

The agent has two controllers sharing spatial state (Figure 2). The *reactive controller* runs continuously, handling perception, mapping, and short-horizon motion; it builds on ASCENT (Gong et al. 2026), a value-map controller in the VLFM (Yokoyama et al. 2024) family, augmented with an online per-detection VLM verification gate and a triggered slow-thinking brain. A *deliberative controller* runs only when a triggering event fires: it then executes a bounded reason-retrieve-act loop (Yao et al. 2023) and returns a typed command the reactive controller executes. The two couple at the data level: the reactive controller stores what it observes, and the deliberative controller turns selected entries into typed navigation targets.

Three independent switches set the deliberative layer’s behaviour: *when* it fires, *how* its evidence budget is spent, and *which* memory-dependent capabilities are enabled. Section 4.3 turns each off in turn.

3.2 Reactive Controller

The reactive controller maintains, per floor, four 2D maps built by projecting depth into the odometry frame. An *occupancy map* supports collision-free motion and frontier extraction. A *value map* stores in every free cell c the target-conditioned score $V(c) = \text{sim}(\phi_{\text{img}}(I_c), \phi_{\text{txt}}(g))$, with $\phi_{\text{img}}, \phi_{\text{txt}}$ the encoders of a pretrained image-text matching model (Li et al. 2023) and I_c the view associated with c . A *semantic grid* (Section 3.3) records *what* was seen and *where*; the reactive pipeline populates it from quantities it already computes—no extra model call—and the deliberative controller alone consumes it. An *object map* accumulates the

goal detections that pass the gate below—the map that can authorize stopping. Frontier points on the boundary with the unknown, read off the occupancy map, give the candidate exploration targets.

Goal candidates come from a detection ensemble (Peng et al. 2024; Liu et al. 2024; Zhang et al. 2024) and a lightweight segmenter (Zhang et al. 2023), room labels from a scene classifier (Zhou et al. 2018) and pixel semantics from an RGB-D network (Jiang et al. 2018); a pretrained point-goal policy (Wijmans et al. 2020) executes motion toward a chosen target, and inter-floor transitions are handled by a stair finite-state machine. Every candidate detection then passes an *online per-detection verification gate* before it is fused into the object map. The gate shows a VLM the current frame cropped to the candidate box together with the proposed category, and returns an accept/reject decision; rejected candidates are discarded and may be proposed again from a later frame. Because a fused cloud merges detections by category, no individual detection can be removed once fused; verifying first keeps rejected candidates out of the map that authorizes stopping, whereas a later check could only mark them as suspect.

3.3 Coordinate-Anchored Memory

The reactive controller stores room labels, confirmed objects, and pose-tagged keyframes in two coordinate-indexed structures, shown architecturally in Figure 2 and on an example in Figure 3(a).

Semantic grid We maintain a coarse grid whose cell $\Pi(\mathbf{x}) = \lfloor \Pi_{\text{px}}(\mathbf{x})/8 \rfloor$ aggregates an 8×8 block of occupancy pixels, reusing the occupancy map’s own world $\rightarrow$ pixel transform Π_{px} to guarantee consistency. Each cell stores (i) a room-type vote histogram, accumulated rather than overwritten so that transient misclassifications average out; (ii) a bounded ring of *keyframe indices* whose view frusta covered the cell; and (iii) a sparse table of *object instances* (class, back-projected world position, observation count, supporting keyframes). These are context objects—categories other than the goal, which the reactive controller tracks separately in its object map—paired with the views that support them, so that deliberation can retrieve them by location. We call an instance *confirmed*, and make it retrievable, once it has been observed at least twice, which suppresses spurious one-shot detections. At 0.4 m cells, retrieval keyed by world coordinates tolerates pose jitter; because every cell and keyframe also carries its floor index, a query can reach floors the robot has left.

Keyframe store In parallel we keep a bounded store of low-resolution keyframe thumbnails, sampled on translation/rotation thresholds, each tagged with its pose, floor, and step index and back-filled with the room label and visible objects. Eviction enforces a *per-floor* capacity, so a newly entered floor cannot evict the memory of earlier ones.

Retrieval interface Given a query world position, the memory returns four things: the local room estimate and its confidence, the nearby confirmed instances, a coverage statistic (how much of this neighborhood has actually been seen),

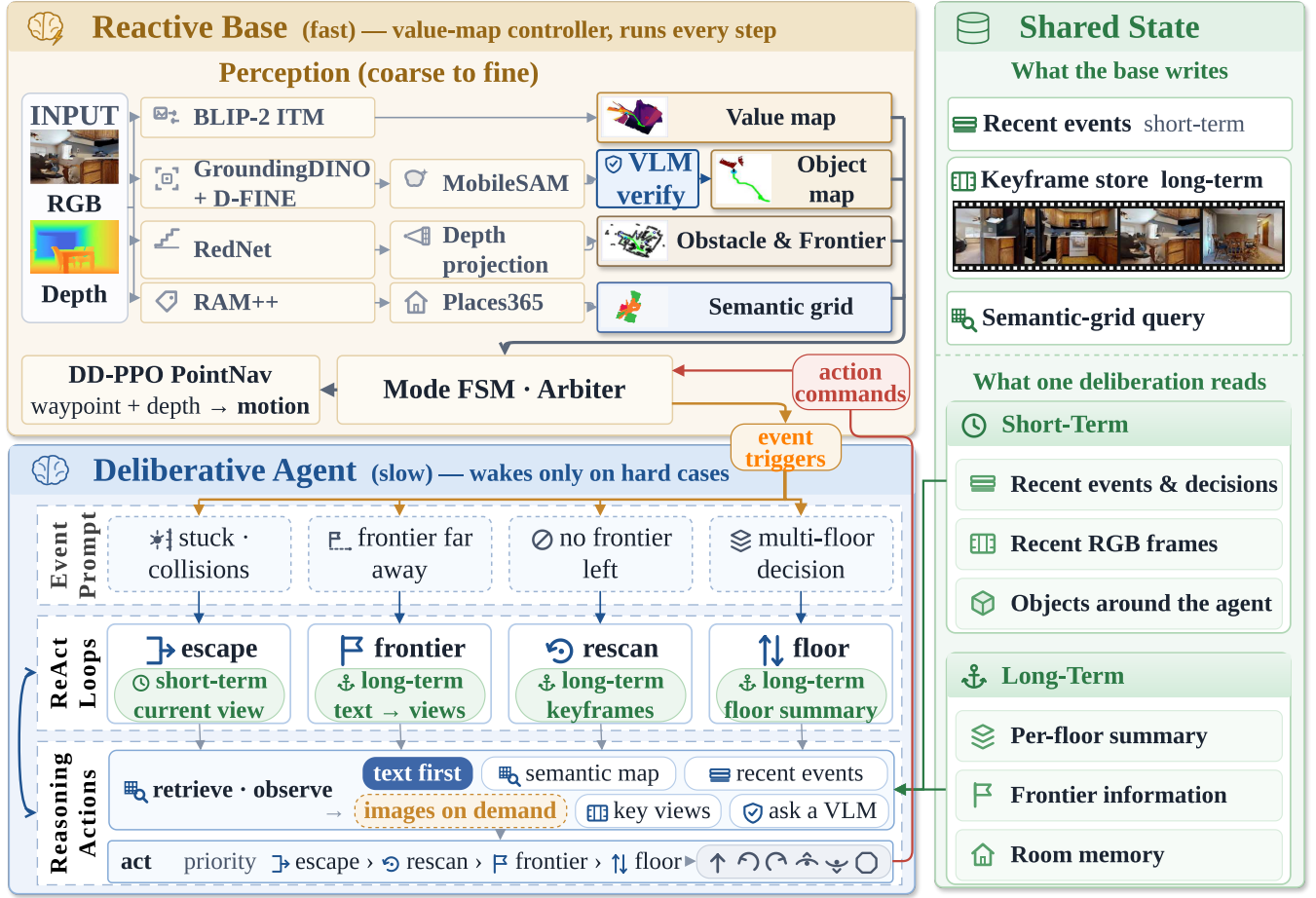


Figure 2: System architecture. The always-on *Reactive Base* (top, fast) perceives, maps, and moves at every step and writes the shared state (right)—short-term event context plus the coordinate-anchored memory itself, the semantic grid and the keyframe store. An event-triggered *deliberative agent* (bottom, slow) wakes only at a structural impasse, reads that memory in a bounded ReAct loop—annotated with the horizon it queries—and returns typed commands under strict budgets. Any deliberative failure degrades back to the Reactive Base, which executes every motion primitive.

and a ranked set of candidate keyframes. Candidates are the cell’s view-footprints together with the supporting views of nearby instances, passed through a field-of-view gate and a distance band and ranked by $s(k) = \cos(\Delta\beta_k) - \lambda d_k$ subject to $|\Delta\beta_k| \leq \frac{1}{2} \text{HFOV}$, with $\Delta\beta_k$ the bearing offset of keyframe k from the query and d_k its distance. Image recall follows a strict fallback ladder—exact footprint hit, relaxed footprint search, the frontier’s discovery frame, and finally an empty result telling the caller to proceed text-only. A retrieval never interrupts control: when no view can be recalled the loop continues on a step-indexed context and then a floor-level summary.

3.4 Event-Triggered Deliberation

Four structural impasses wake the deliberative controller, each detected from quantities the reactive controller already maintains, and each answered by its own bounded loop. Every invocation obeys a per-invocation budget—at most $M=3$ reasoning rounds, at most two images in any one call, and a

wall-clock cap—on top of the per-run budget of Section 3.5.

Contact stalls From the platform’s cumulative collision count n_t we form a per-step contact indicator over a sliding window W and call the robot stuck when $\sum_{\tau=t-W+1}^t \mathbf{1}[n_\tau > n_{\tau-1}] \geq \kappa$, with a rising-edge latch and a cooldown so one physical stall yields one event; a no-net-displacement odometry test over a longer window is a secondary signal. Both are suspended while the controller is already deliberating, so a deliberate pause is not mistaken for a stall. Recovery is then tiered, so that the common stall costs no model call at all. The first tier suppresses the pursued frontier target and forces a replan, which resolves stalls caused by an unreachable or ill-chosen target. Only if the robot fails to move away within a fixed displacement window, or stalls again under the suppression, does the controller escalate: a VLM is queried with the current view for an escape heading, and the matching short macro-action is executed. Sufficient net motion ends the loop; otherwise the query repeats with previously tried directions as explicit negatives, up to the

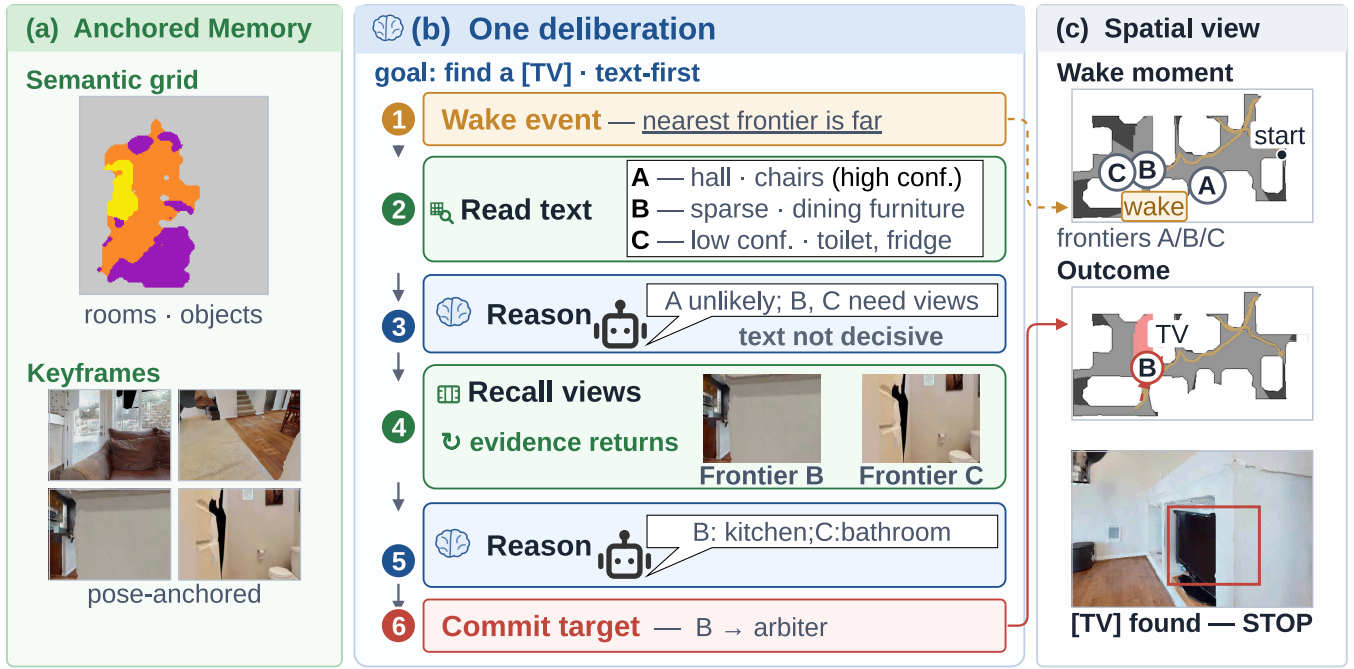


Figure 3: One deliberation, end to end, on one HM3D evaluation run (goal: TV, success). (a) Anchored Memory—a semantic grid, tinted by per-cell room type (here hall in orange, kitchen in purple, dining room in yellow, over light grey for space not yet observed), and pose-anchored keyframes—is written by the Reactive Base as it explores and reset at the start of every run. (b) The nearest frontier lies far, so the bounded reason–retrieve–act loop wakes and reads textual memory first. When that round cannot separate frontiers B and C, it retrieves one stored view toward each—remembered, not re-visited—submits those images in a second round, and commits a single typed target. (c) The wake moment in space: trajectory (tan), wake pose, and frontiers A/B/C; B opens onto a kitchen–living area, and driving there reveals the TV (red box).

round budget. Exhaustion disables the offending region and falls through to the reactive base’s own recovery.

Distant frontiers Committing to a far frontier costs dozens of steps to undo, and a similarity heatmap alone offers no way to tell two distant frontiers apart. With frontier candidates $F = \{f_i\}$ and robot position $\mathbf{p}$, the controller deliberates when even the nearest candidate is beyond a distance gate, $\min_i \|\mathbf{f}_i - \mathbf{p}\| > d_{\text{far}}$, subject to the same cooldown and budget checks; near frontiers keep the cheap reactive path. The loop acquires *visual evidence only when text is insufficient* (Figure 3). For each of the nearest far candidates it assembles a textual description from the semantic memory (room estimate, nearby objects) and the target’s room affinities, read from a fixed category–room knowledge graph. The first round is *text only*: the model either commits to a candidate or, if the descriptions are too ambiguous, requests first-person images for at most two named candidates. Only then does the controller retrieve the best-available keyframe per requested candidate by $s(k)$ and issue a second, image-bearing round, which turns a fixed image budget into adaptive, per-decision expenditure. The chosen frontier becomes a persistent deliberative target (Section 3.5); a parsing failure or exhausted budget returns control to the reactive value-ranking path.

Frontier exhaustion When the frontier set empties before the goal is found, the reactive base has nothing left

to explore and would end the run—exactly when a missed opening is most consequential and what the robot has already seen is most informative. Rather than accept that stop, the controller re-examines the current floor’s stored keyframes—remembered observations, not a physical re-traverse—subsampling uniformly into one indexed montage, and asks in a single image-bearing call whether any remembered view shows an overlooked passage (staircase, door, opening) into unexplored space. An identified view turns its anchored pose into a persistent goal with a validity horizon, and exploration targets previously suppressed near that pose are re-enabled so arriving there can regenerate frontiers. If the model declines, the reply is unparseable, or the budget gate is closed, the run ends exactly as it would have without rescan. Rescan is capped per floor and draws from the shared per-run budget.

Floor changes When exploration on the current floor stalls and other floors are known, the evidence that settles the choice sits on floors the robot has left. The controller sends a purely textual per-floor summary to the VLM service to decide whether to change floors, again fail-open.

3.5 Arbitration and Budget Control

A single arbiter reconciles reactive motion with deliberative commands under a fixed priority order: a recovery macro-action preempts everything; the stair finite-state machine re-

Method	HM3D v1		MP3D	
	SR $\uparrow$	SPL $\uparrow$	SR $\uparrow$	SPL $\uparrow$
ZSON (Majumdar et al. 2022)	25.5	12.6	15.3	4.8
ESC (Zhou et al. 2023)	39.2	22.3	28.7	14.2
L3MVN (Yu, Kasaei, and Cao 2023)	50.4	23.1	34.9	14.5
VLFM (Yokoyama et al. 2024)	52.5	30.4	36.4	17.5
SG-Nav (Yin et al. 2024)	54.0	24.9	40.2	16.0
ASCENT (Gong et al. 2026)	65.4	33.5	44.5	15.5
Ours (Full)	68.75	33.97	47.29	18.01

Table 1: Ours leads every zero-shot baseline on both benchmarks and on both metrics, so the added successes do not come from longer paths.

tains control during inter-floor transitions; otherwise navigation proceeds toward a detected goal or a deliberative target. Targets from the distant-frontier loop or rescan are *persistent*: the arbiter suppresses cheap reactive preemption until the target is reached or its validity horizon elapses, so the controller deliberates once and then commits rather than thrashing.

A *model call* is any invocation of the external VLM service by the deliberative controller, and *image-bearing* calls are the subset carrying one or more images; the verification gate and the floor decision call the same service under caps of their own and are counted separately. Calls are bounded at two levels: the per-invocation caps above, and a *per-run* cap of 20 model calls with a tighter cap of 10 image-bearing calls, shared across all deliberative loops. A per-step watchdog degrades the controller for the rest of the run if any single decision exceeds 180 s, and a circuit-breaker disables model calls after consecutive failures. Degradation is deterministic rather than best-effort: every failure path—empty memory, service outage, unparsable reply, exhausted budget—has the reactive base as its single terminal state, so no deliberative failure can leave the agent without a controller. Table A1 in the appendix summarizes modes, triggers, and budgets.

4 Experiments

4.1 Benchmarks and Metrics

Benchmarks We evaluate on the ObjectNav *val* splits of HM3D v1 (Ramakrishnan et al. 2021) and MP3D (Chang et al. 2017) in the Habitat simulator (Savva et al. 2019). Both are built from photorealistic scans of real buildings, so the robot moves through the clutter and sight lines of houses that exist. HM3D v1 *val* contains 2000 episodes across 20 buildings and six goal categories (chair, bed, plant, toilet, sofa, TV monitor). MP3D *val* contains 2195 episodes across 11 buildings and 21 categories, among them small and easily confused targets such as cushion, towel, and chest of drawers; its buildings are larger and more often span several floors.

Metrics Success Rate (SR) is the fraction of episodes in which the robot calls stop within 1.0 m of an instance of the goal category, from a pose where that instance can be viewed. Success weighted by Path Length (SPL) (Batra et al. 2020) scales each success by the ratio of the shortest available path

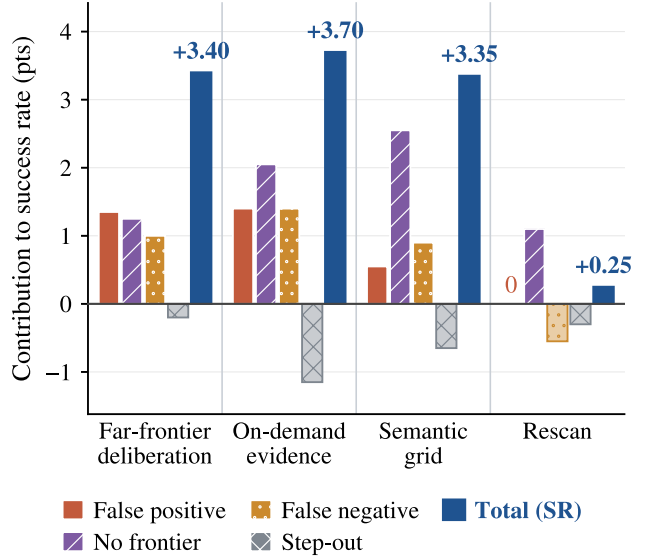


Figure 4: Failure-class decomposition of each ablation gap on HM3D. Each group is one component, measured by the Table 2 row that takes it away; within a group, the four coloured bars are that component’s contribution in SR points to each failure class and they sum to the blue **Total**, while bars below zero run the other way. Each component signs the spectrum differently: deliberation spreads its 3.40 points over the three decision-error classes, the semantic grid concentrates 2.55 of its 3.35 in no-frontier stops, and on-demand evidence (3.70) leads on both detection errors.

to the path actually walked, and scores zero on failure—SR asks whether the object was found, SPL asks what the search cost. An episode that reaches 500 steps ends in failure.

4.2 Main Results

Ours (Full) leads every zero-shot baseline in Table 1 on both metrics and on both benchmarks: 68.75% SR and 33.97% SPL on HM3D v1 *val*, 47.29% SR and 18.01% SPL on MP3D *val*. Ours (Full) averages 5.00 deliberative model calls and 205 environment steps per episode.

4.3 Ablations

Table 2 reports five switch-level ablations and Figure 5 plots success against call count, every arm on the same 2000 episodes as Table 1. Greedy Frontier—choosing among far frontiers by argmax instead of deliberating over them—costs 3.40 SR points, with a paired bootstrap 95% confidence interval of [1.70, 5.05] and McNemar $\chi^2=15.06$ ($p < 0.001$).

Always-Deliberate holds every capability fixed and removes only the trigger, waking the loop at every frontier choice rather than at the far ones. It reaches 64.80% SR—below Greedy Frontier, which never deliberates over a frontier at all—costing 3.95 SR points against the full system ([2.20, 5.65]). It is given strictly more of the model than the full system—more calls per episode, and more of them image-bearing—and returns less success for it. Access to the

Configuration	SR $\uparrow$	SPL $\uparrow$	Calls	Img	Think (s)
Ours (Full)	68.75	33.97	5.00	3.04	17.2
Greedy Frontier	65.35	32.78	1.49	1.49	5.7
Always-Deliberate	64.80	31.36	8.87	4.88	24.3
One-Shot	65.05	31.80	3.75	3.75	17.4
w/o Semantic Grid	65.40	32.49	4.99	3.02	18.5
w/o Rescan	68.50	33.92	5.00	3.04	17.2

Table 2: Deliberation earns its gain through *when* it fires and *what* it reads: firing only at structural impasses leads both on SR and on SPL, while consulting the model at every frontier choice (Always-Deliberate) scores below letting the value map choose (Greedy Frontier). Each row changes one switch of the full system: Greedy Frontier and Always-Deliberate change *when* deliberation fires, One-Shot *how* the same image budget is used, w/o Semantic Grid *what* it may read, and w/o Rescan *when* to stop. Per episode, *Calls* counts the deliberative controller’s model calls and *Img* how many of those calls carried images—the verification gate and the floor decision run on budgets of their own—and *Think* is the deliberation time.

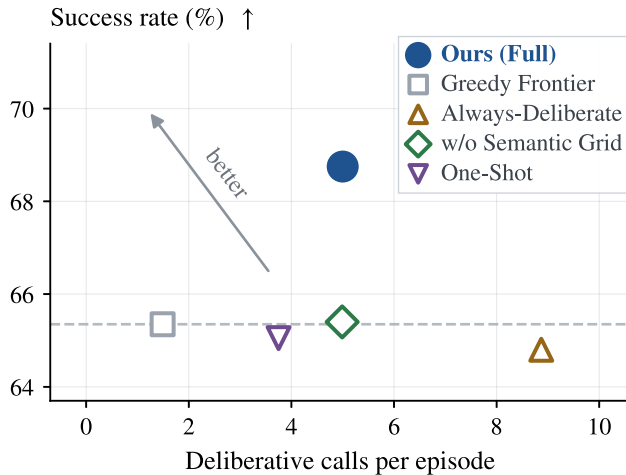


Figure 5: The gain is not bought with deliberative calls. Always-Deliberate makes the most calls per episode and scores lowest; One-Shot makes fewer than Ours and lands on the dashed line—the Greedy Frontier success rate—as does w/o Semantic Grid, which makes as many calls as Ours. Only the full system, which fires at structural impasses *and* reads the anchored memory when it does, sits above that line. Up and to the left is better, exact values in Table 2.

model is therefore not what produces the gain; the policy that decides when to spend it is. One-Shot keeps the trigger policy fixed but supplies the images in one full-context query, isolating adaptive evidence acquisition: it costs 3.70 SR points ([2.00, 5.45]) while making *fewer* paid calls than the full system, every one of them image-bearing. What that row gives up is buying evidence on demand, not access to imagery. w/o Semantic Grid takes away the abstract tier alone—room types and confirmed instances—and leaves the literal tier

standing: keyframes are still written, and the loop still recalls them. It spends the same budget as the full system on both axes (4.99 against 5.00 calls per episode, 3.02 against 3.04 of them image-bearing) and still costs 3.35 points ([1.75, 5.00]). Neither fewer calls nor less imagery can explain that: one read surface is the only thing that changed.

4.4 Analysis

Rescan attribution Rescan converts stored views into new navigation targets when the frontier is exhausted: it fires on 6.1% of episodes, 41.4% of which yield a new goal from a remembered view.

Verification gate The gate changes what the persistent map contains rather than merely flagging doubtful entries: it vetoes 16.7% of candidate detections before fusion, so they never reach the object map.

Where the gain comes from Every episode ends in success or in exactly one of four failure classes, taken from its recorded termination state rather than hand-labelled: two detection errors (false positive, false negative) and two termination outcomes (no frontier left, step-out). The per-class differences between an arm and the full system therefore sum to their SR difference, and Figure 4 uses that identity to attribute the gain. Deliberation returns 1.35, 1.25, and 1.00 points from false positives, no-frontier stops, and false negatives—three classes at once, no single one carrying the result. The semantic grid lands in one: 2.55 of its 3.35 points come from the premature stops it is built to prevent. The two leave different signatures, and are complementary rather than redundant. An episode kept alive by suppressing a premature stop can only end in success or at the step cap, and the split favours success: of the 3.60 points deliberation redirects, 3.40 arrive as success.

5 Conclusion

We presented a hierarchical fast-slow agent for zero-shot object-goal navigation: an always-on reactive value-map controller writes a coordinate-anchored memory, and an event-triggered, budget-bounded ReAct agent reads it, reasoning from text first and retrieving imagery only when text cannot decide. A pre-fusion gate keeps rejected detections out of the map that authorizes stopping, and every deliberative failure returns control to the reactive controller. It reaches 68.75% SR on HM3D v1 val and 47.29% on MP3D val, leading every zero-shot baseline of Table 1; choosing among far frontiers by argmax instead costs 3.40 SR points. What the robot has already seen is the evidence worth reasoning over—and keeping that reasoning bounded and fail-open is what makes it deployable.

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Mode	Trigger	Per-invocation budget
Recovery (tiered)	contact accrual, or the secondary no-displacement test (Section 3.4)	tier 0: replan, no call; model tier: ≤ 3 rounds, ≤ 2 images in any one call
Distant frontier	nearest frontier beyond the distance gate (Section 3.4)	≤ 3 rounds, ≤ 2 images in any one call
Inter-floor	floor stalled, multiple floors known	single text call
Rescan	frontier set empty before reactive stop	single montage call, capped per floor

Per run: ≤ 20 model calls, of which ≤ 10 may be image-bearing; ≥ 15 steps between re-triggers of the same loop; rescan ≤ 2 per floor over a montage of ≤ 8 keyframes; a per-step watchdog disables model calls for the rest of the run if any single decision exceeds 180 s, and a circuit breaker disables them after consecutive service failures.

Table A1: Deliberation modes, triggering conditions, and the structure of their budgets.

A Implementation of the Deliberative Loops

Each loop of Table A1 is implemented as a declarative skill specification—a Markdown file whose frontmatter names the loop’s triggering events, its model backend, its per-invocation budget, and the schema of the typed command it must return—loaded at startup and dispatched by a single runtime. Loops are therefore added, disabled, or re-budgeted without touching the reactive base, and the trigger and budget columns of Table A1 are read directly from those files rather than restated in code. A Python registry provides a fallback implementation for any loop whose specification fails to load, which is one of the paths that degrades to the reactive base.

B Computing Infrastructure

All experiments run on a workstation with a single NVIDIA GeForce RTX 3090 GPU (24 GB), an Intel Core i7-14700KF CPU, and 32 GB of RAM, under Ubuntu 20.04. The software stack is Python 3.9, PyTorch 2.1.0 with CUDA 11.8, Habitat-Sim and Habitat-Lab 0.3.1, and ROS Noetic. The perception models of the reactive base—the image-text matching model, the open-vocabulary detection pipeline, the segmentation and room-classification networks, and the point-goal policy—run locally on the GPU; deliberation and the verification gate invoke the external VLM service over its API, under the budgets of Table A1.

Anchored semantic grid



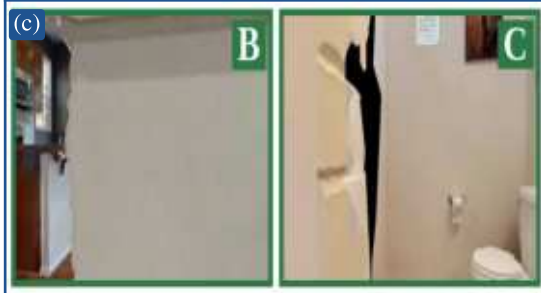
room-type votes and confirmed instances

Pose-anchored keyframe store



thumbnails written by the reactive base, no extra model call

What recall_view returned



text was inconclusive; ≤ 2 FOV-gated views fetched

Outcome



committed frontier, then the goal in view

Figure A1: One deliberation on one HM3D evaluation run (goal: TV, success—the run of Figure 3), at the distant-frontier trigger. (a) the coordinate-anchored semantic grid the reactive base had built by the wake moment. Cells are tinted by their winning room-type vote—on this floor hall (orange), kitchen (purple), and dining room (yellow)—over light grey for space not yet observed. (b) the pose-anchored keyframe store at that moment; the two views the loop went on to request are outlined and labelled **B** and **C**. (c) the same two views as `recall_view` returned them, one per contested frontier, after text alone could not separate the candidates. (d) the committed frontier on the occupancy map, and the frame in which the goal came into view.